Assembly instruction

1. Loom assembly

Prepare all materials

1. Set up the printer following the printer manual, leave the nozzle and LCD screen off.

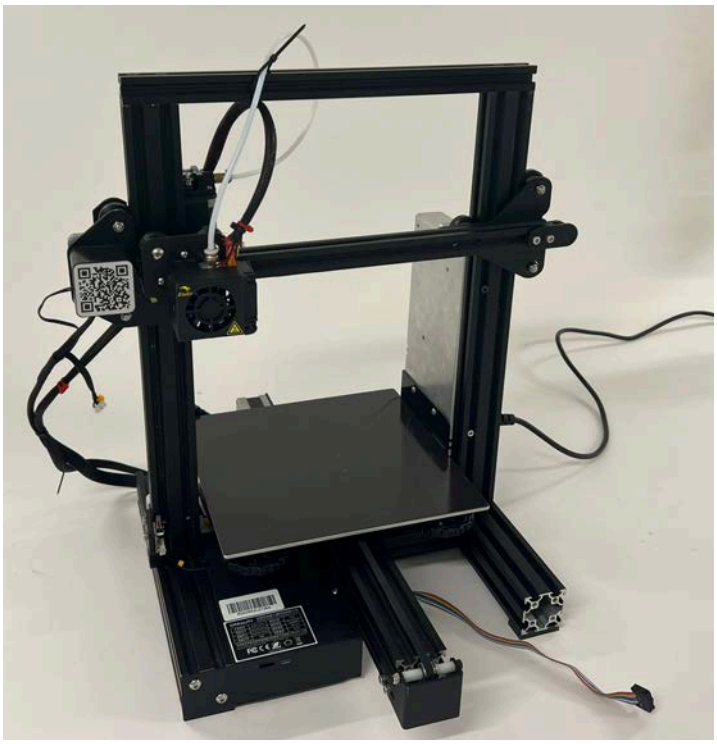


Figure 1. The Ender 3D printer without nozzle and LCD screen connected.

2. Prepare all materials according to the list of materials.

3. Bend the reed rod and attach the reed onto the rod. (Sizes for each part starting at the bottom left: 2.5cm, 10cm, 10cm, 30cm, 10cm, 10cm, 2.5cm) (angles for one side are are 90 degrees sideways, 130 degrees forward, 90 degrees sideways opposite to the first)

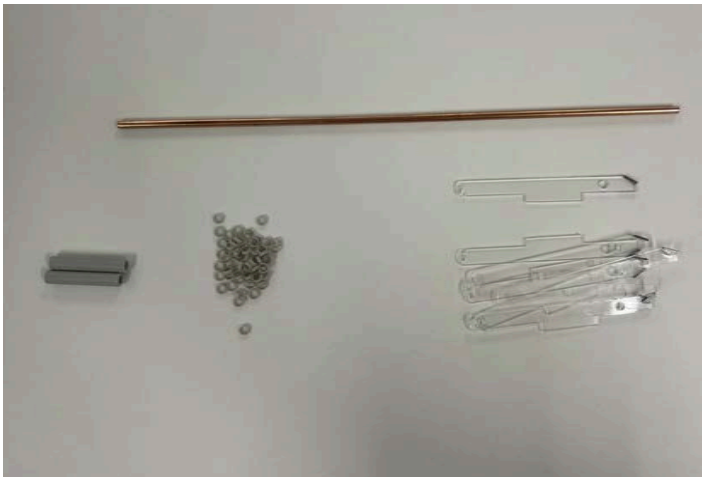
Figure 2. Reed.

4. Glue the nail in the nozzle (make sure it sticks out 2 cm) and glue the magnets to the front of the magnet beam.

Figure 3. Nozzle with nail.

Figure 4. Magnet beam.

5. Attach the steel tips to the end of the levers with a strong glue. Take the steel rod and alternate the levers and spacers, finish both sides with the big spacer.



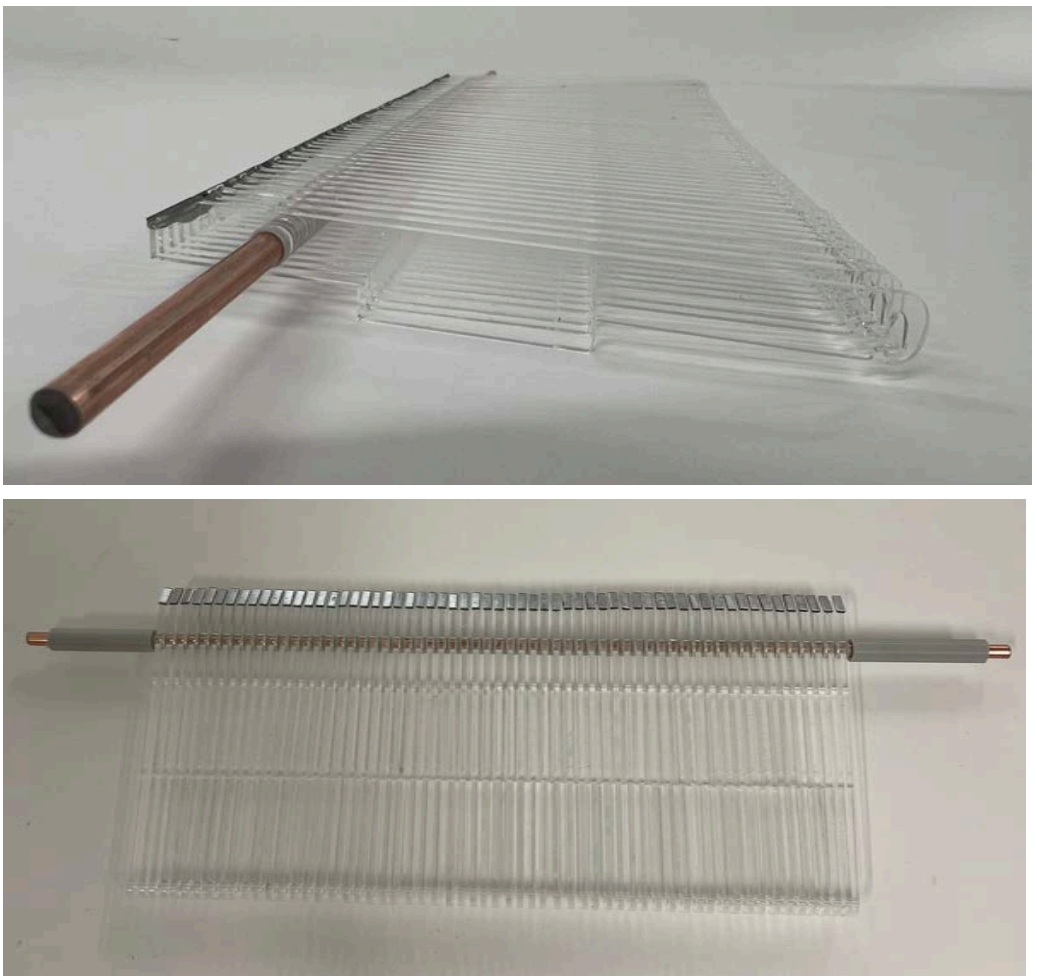


Figure 5. Rod with levers.

6. Add the Apron chords to the warp and cloth beams if you like to have these by screwing four holes in the warp beam (locations: 3 cm, 12cm, 21cm, and 30cm).

Figure 6. Apron chords connected to warp/cloth beam.

Assembly

Figure 7 shows the assembly connections with annotations of all mounting points of each component. This image can be used to find the locations of each component.

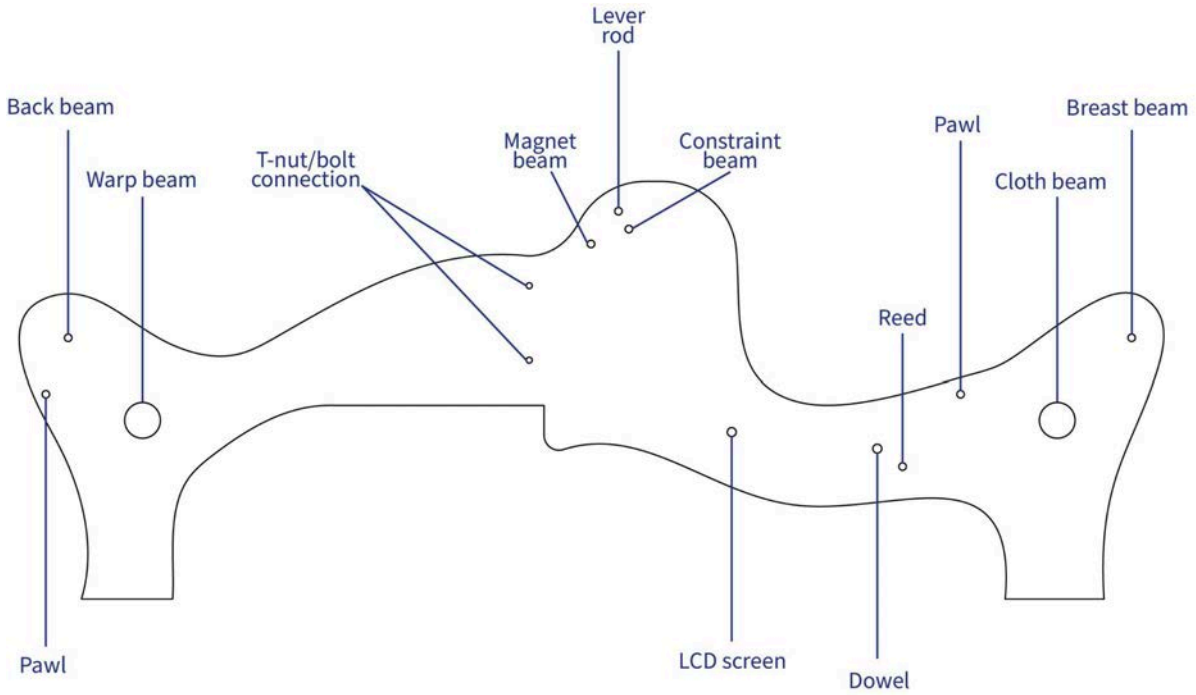


Figure 7. Overview of the inside of the right side panel with indication on where to connect all parts.

7. Connect the screws and t-nuts to the side panels, as shown on the image. The t-nut should go on the inside of the right side panel.

Figure 8. Bolt and t-nuts connected to the side panel.

8. Now you can attach all rods and beams (except the rod with levers) to the right side panel (see locations in figure 7). Connect the side panel to the right side of the printer (when looking at the front of the printer), with the t-nut connections.

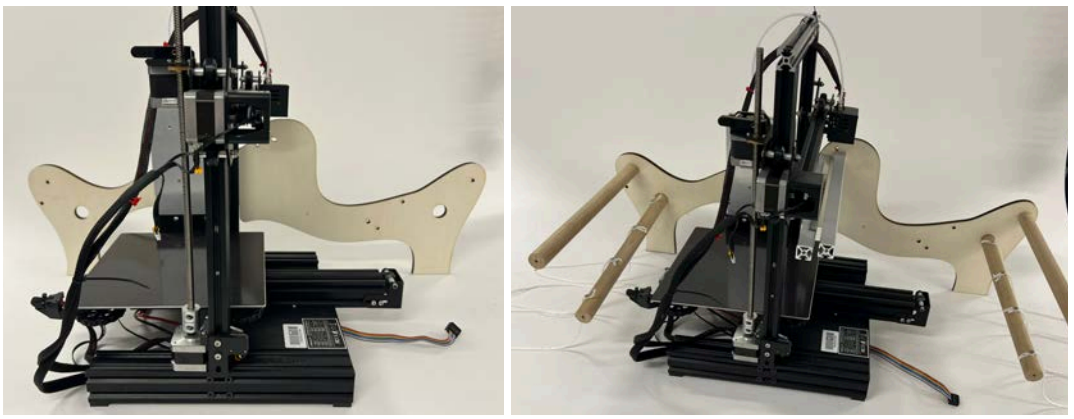


Figure 9. Right side panel connected to the printer, with rods and beams connected.

9. Add the rod with levers in place. Then place the left side to the printer and add the rod with levers in between both panels.

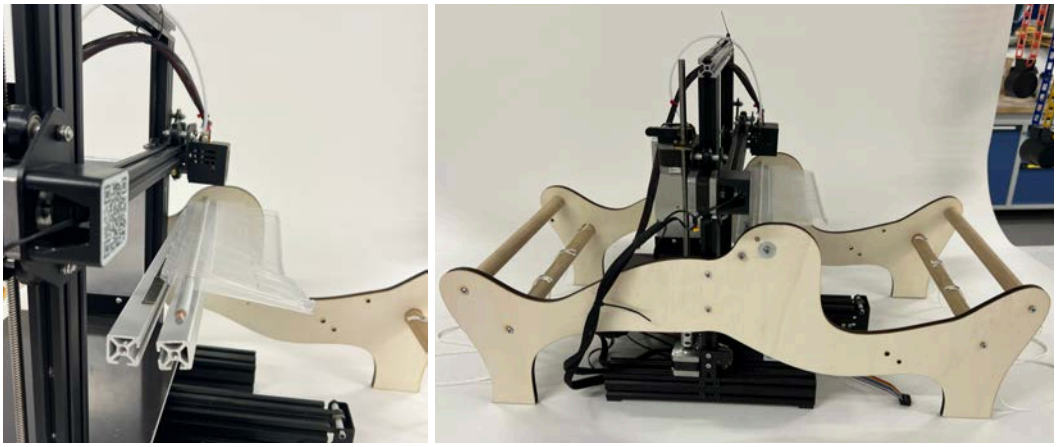


Figure 10. Rod with levers is connected and the left side panel is connected.

10. When all rods and beams are in the right location, fasten the screws, including the screws of the side panels. Make sure to check the height of the side panels to make them even.

11. Screw the ratchets to the warp and cloth beam. Then connect the pawls with a bolt and nut in the holes. Make sure the gear lock fits over the gear and can move freely.

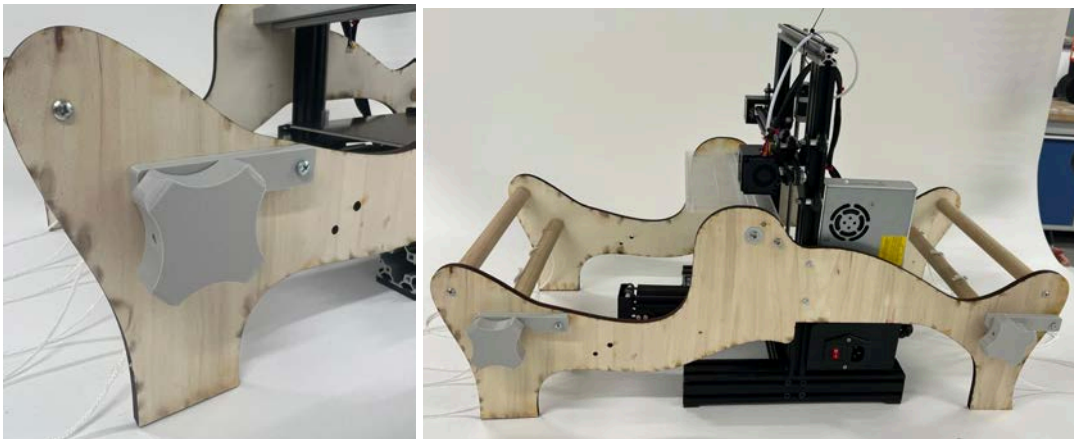


Figure 11. Ratchet and pawl connected.

12. Screw the block onto its position and screw the screen to the block.
    PLAATJE block and screen.

Figure 12. LCD block connected to the right side panel.

13. Glue the dowels to the side panels and add the reed.

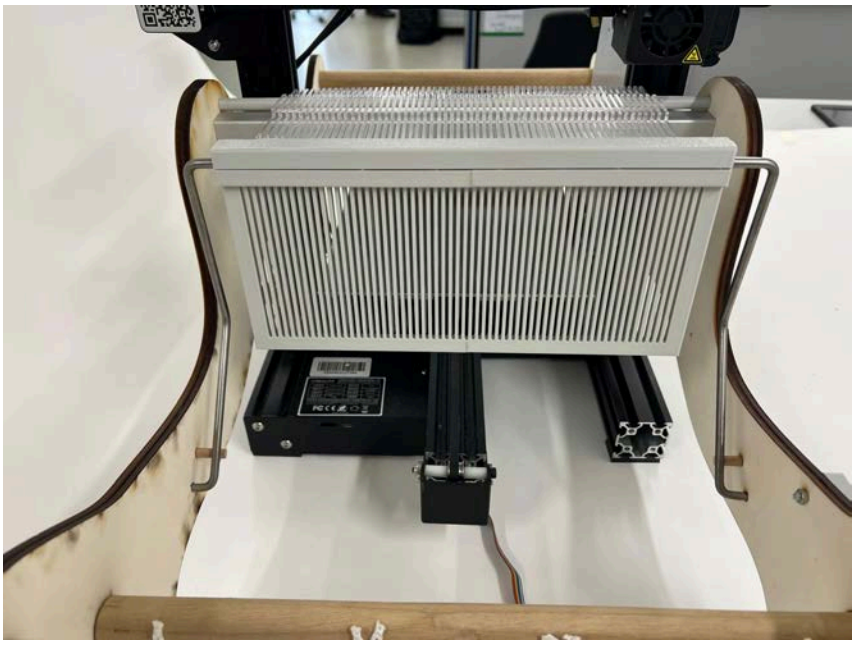


Figure 13. Dowels and reed.

14. Screw the nozzle in place.

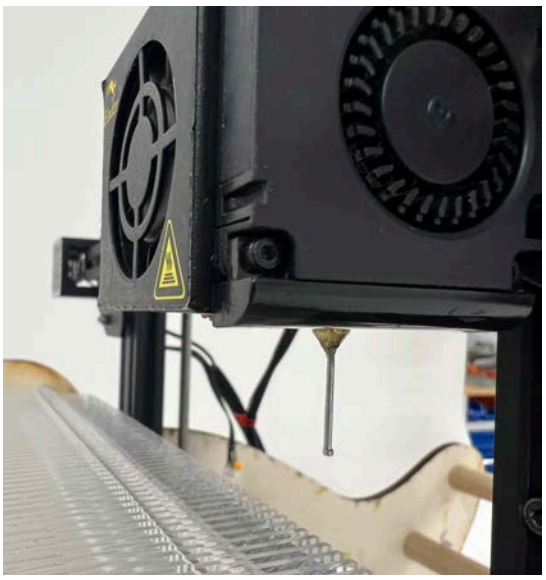


Figure 14. Nozzle connected to the printhead.

15. Attach all heddles to the levers

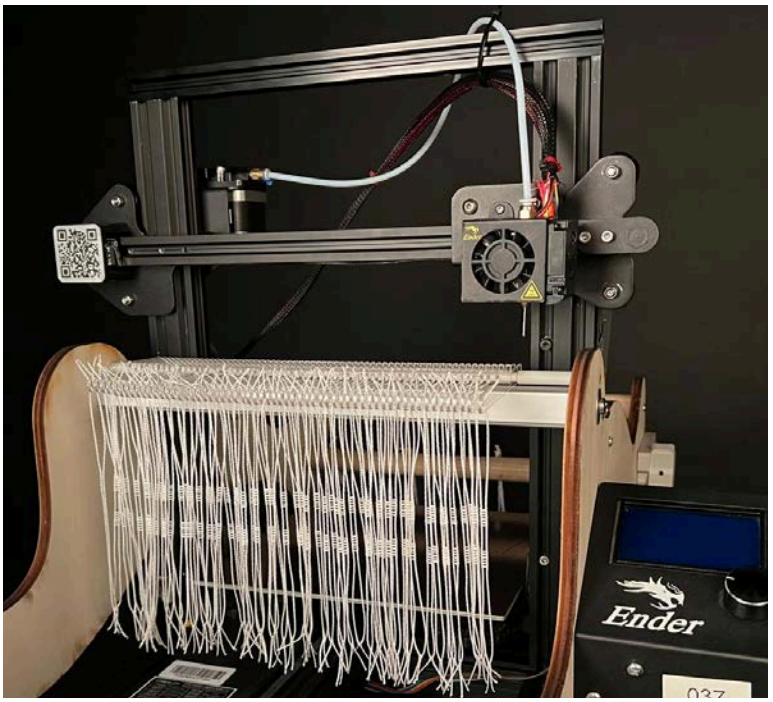


Figure 15. Attached heddles to the levers.

16. Connect the warp threads like any other loom.

17. The loom is now ready for weaving!
    (Make sure to check the print velocity and acceleration and set X and Z to max for faster selection.)

2. Conversion tool set up

1. Install Rhino
2. Download the following Grasshopper plugin:
    - Human UI [15]
3. Open the grasshopper file, rhino and grasshopper will automatically start.
4. A user interface window will automatically popup. The window explains the four steps to convert your .wif file to a .gcode file to use it for weaving on the loom.

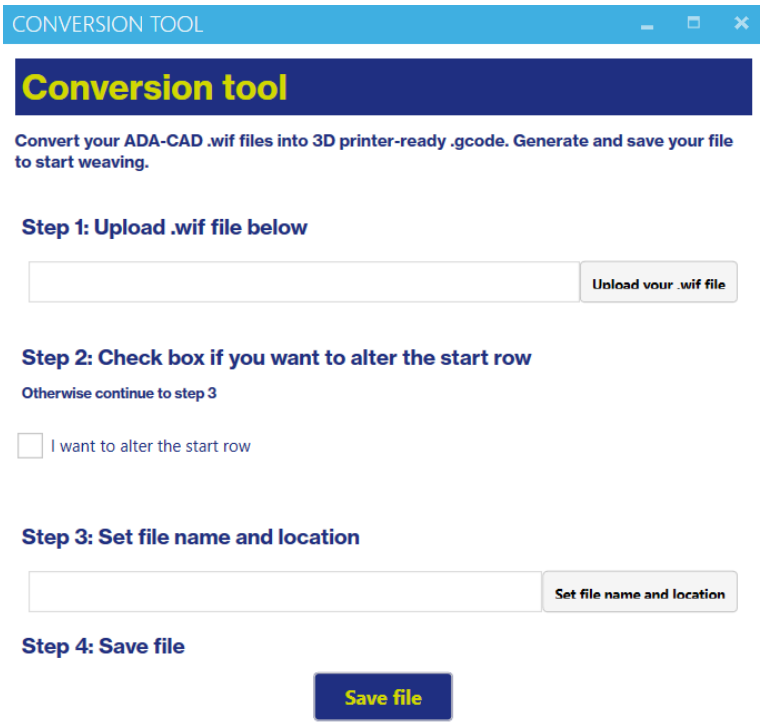


Figure 16. Conversion tool user interface.

3. Disassembly loom

- To disassemble the loom add-on from the printer, first unscrew the LCD screen and remove the nozzle.
- Then unscrew all rods and beams on the left side. When all of the screws are removed you can unscrew the bolts that connect the panel to the printer. You can now take off the left side panel and the rod with levers.
- If you want to keep the warp threads on the beams, you can unscrew the gear handles of the beams and take them off. If not you can now remove the warp threads.
- Unscrew the bolts that connect the right panel to the printer and and take off the right side panel with the rods and beams still connected.
- You can unscrew the side panel screws connecting the left panel to the printer and take off the panel with all rods and beams still connected. (You can also unscrew all rods and beams for easier storage.)
- Insert a new nozzle and screw the LCD back into its original location and you can print again. (Make sure to check the printing speed and change back to original speeds).